

WASP-10b

R-1269

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_191102 · created 2026-07-31T13:48:44+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458789.7416 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.049
Transit depth (Rp/Rs)^2	2.45 +/- 1.52 [%]
Orbital Inclination (inc)	88.56 +/- 1.63
Airmass coefficient 1 (a1)	0.963 +/- 0.019
Airmass coefficient 2 (a2)	0.034 +/- 0.063
Scatter in the residuals of the lightcurve fit is	1.98 %
Best Comparison Star	#3 - [259, 166]
Optimal Aperture	3.77
Optimal Annulus	6.66
Transit Duration (day)	0.088 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.049 vs 0.1592 (deviation 0.07σ)
Duration fit vs published	0.088 d vs 0.0946 d (deviation 0.6σ)
Mid-time O–C	6.3 min
Depth SNR	3.2
Residual scatter	1.98 %

Light curve

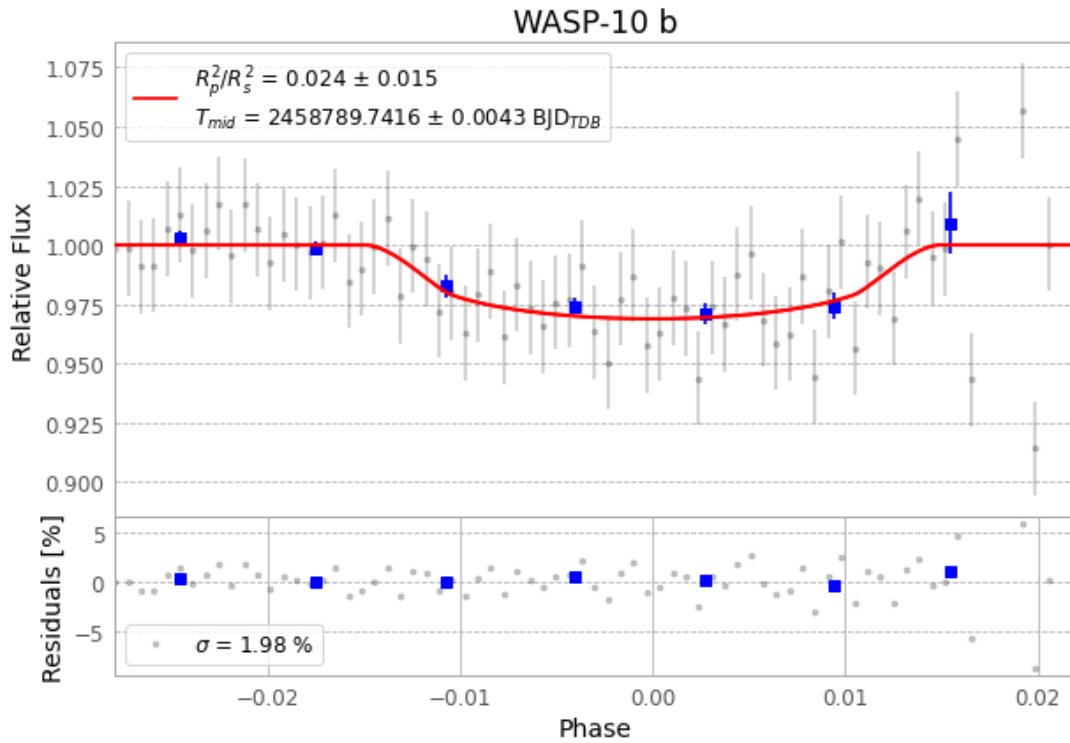


Figure 1 — FinalLightCurve_WASP-10 b_2019-11-01.png (SHA-256 97951f6e90493b00...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [261, 266], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f0c7c5ada769d59d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

84 FITS frames (55 MB), WASP-10191102033612.FITS → WASP-10191102074512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-191102.log (86ec9a222902...), FinalLightCurve_WASP-10 b_2019-11-01.png (97951f6e9049...), FinalLightCurve_WASP-10 b_2019-11-01.csv (8b23db10fa0f...)

Compute resources

Wall time 450.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 13:48Z.